

Covid_19_Final_Project

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R Markdown

```
knitr::opts_chunk$set(echo = TRUE)
library(tidyverse)
library(lubridate)
library(dplyr)
```

Getting Covid 19 Data

Begin by reading in 4 data sets from John Hopkins Covid 19 Github website

```
global_cases <- read.csv(
  "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_time_series/csse_covid_19_time_series/global_cases_time_series.csv"
)

global_deaths <- read.csv(
  "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_time_series/csse_covid_19_time_series/global_deaths_time_series.csv"
)

us_cases <- read.csv(
  "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_time_series/csse_covid_19_time_series/us_cases_time_series.csv"
)

us_deaths <- read.csv(
  "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_time_series/csse_covid_19_time_series/us_deaths_time_series.csv"
)
```

Checking to see if data was properly read in.

```
head(global_cases)
head(global_deaths)
head(us_cases)
head(us_deaths)
```

Data looks good for all four csv files.

Cleaning and organizing data

Cleaning and organizing Global data sets.

I don't need lat/long for my analysis. Each state or country is a row and each date is a column. Will put date, cases, deaths in their own column. Rename Region and State to be more R friendly with and "/" between them instead of a "."

Tidy global_cases

```
global_cases <- global_cases %>%
  pivot_longer(
    cols = -c('Province.State', 'Country.Region', Lat, Long),
    names_to = "date",
    values_to = "cases"
  ) %>%
  select(-Lat, -Long)

global_deaths <- global_deaths %>%
  pivot_longer(
    cols = -c('Province.State', 'Country.Region', Lat, Long),
    names_to = "date",
    values_to = "deaths"
  ) %>%
  select(-Lat, -Long)
```

Checking global_cases and global_deaths to see if they are formatted the way I want now. Checking to see if data was properly read in.

```
head(global_cases)
head(global_deaths)
```

Data looks good, but we need to change the date format into a date variable. I will do that in the next step as well as combine the global_cases and global_deaths data sets into one called global.

```
names(global_cases)
```

```
## [1] "Province.State" "Country.Region" "date"          "cases"
```

```
names(global_deaths)
```

```
## [1] "Province.State" "Country.Region" "date"          "deaths"
```

Joining global_cases with global deaths. Also changing the date format by getting rid of the "X" in front of the date and replacing periods with /'s.

```
global <- global_cases %>%
  full_join(
    global_deaths,
    by = c("Province.State", "Country.Region", "date")
  ) %>%
```

```

rename(
  `Country_Region` = Country.Region,
  `Province_State` = Province.State
) %>%
mutate(date = mdy(gsub("\\.", "/", sub("^X", "", date))))

```

Checking new global data set.

```
head(global)
```

Data format looks good. Now to look at summary stats for global data set.

```
summary(global)
```

```

##      Province_State      Country_Region      date      cases
## Length :330327      Length :330327      Min. :2020-01-22      Min. :      0
## N.unique : 92      N.unique : 201      1st Qu.:2020-11-02      1st Qu.:     680
## N.blank :226314      N.blank : 0      Median :2021-08-15      Median :    14429
## Min.nchar: 0      Min.nchar: 2      Mean :2021-08-15      Mean :   959384
## Max.nchar: 44      Max.nchar: 32      3rd Qu.:2022-05-28      3rd Qu.:  228517
##                                     Max. :2023-03-09      Max. :103802702
##
##      deaths
## Min. : 0
## 1st Qu.: 3
## Median : 150
## Mean : 13380
## 3rd Qu.: 3032
## Max. :1123836

```

Looks like a large number of dates have zero cases. Lets find how many rows have cases = 0

```
sum(global$cases == 0, na.rm = TRUE)
```

```
## [1] 23500
```

This is a lot, 23500 rows have cases = 0. Filter out rows where cases = 0

```

global <- global %>% filter(cases > 0)
summary(global)

```

```

##      Province_State      Country_Region      date      cases
## Length :306827      Length :306827      Min. :2020-01-22      Min. :      1
## N.unique : 91      N.unique : 201      1st Qu.:2020-12-12      1st Qu.:    1316
## N.blank :210689      N.blank : 0      Median :2021-09-16      Median :    20365
## Min.nchar: 0      Min.nchar: 2      Mean :2021-09-11      Mean :   1032863
## Max.nchar: 44      Max.nchar: 32      3rd Qu.:2022-06-15      3rd Qu.:   271281
##                                     Max. :2023-03-09      Max. :103802702
##
##      deaths
## Min. : 0
## 1st Qu.: 7

```

```
## Median :    214
## Mean   :   14405
## 3rd Qu.:    3665
## Max.   : 1123836
```

We can see the min cases number is 1. So we successfully removed all rows where cases = 0.

Maximum number of cases looks high. Lets check if it looks like a mistake.

```
global %>% filter(cases > 90000000)
```

Ok so we have numerous dates with over 90 million cases so our maximum number of cases of 103 million seems correct.

Cleaning and Organizing United States Data Sets.

Checking column names and content of US data sets.

```
head(us_cases)
head(us_deaths)
names(us_cases)
names(us_deaths)
```

Data looks good. Now I will work with US_cases data set. Start by pivoting so that each date is a row, change date data type into a date data type, and remove Lat and Long_ columns.

```
us_cases <- us_cases %>%
  pivot_longer(cols = -(UID:Combined_Key),
               names_to = "date",
               values_to = "cases") %>%
  select(Admin2:cases) %>%
  mutate(date = mdy(gsub("\\.", "/", sub("^X", "", date)))) %>%
  select(-Lat, -Long_)
head(us_cases)
```

US_cases now has the following columns: Admin2, Province_State, Country_Regions, Combined_Key, date, cases.

Now to tidy the us_deaths dataset. Looking above the column names are the same except that the us_deaths data set has a Population column before the dates.

Selecting columns to use, pivoting so each date column is now a row, and fixing date data type into date data type.

```
us_deaths <- us_deaths %>%
  pivot_longer(cols = -(UID:Population),
               names_to = "date",
               values_to = "deaths") %>%
  select(Admin2:deaths) %>%
  mutate(date = mdy(gsub("\\.", "/", sub("^X", "", date)))) %>%
  select(-Lat, -Long_)
us_deaths
```

Joining us_cases and us_deaths.

```
united_states <- us_cases %>%
  full_join(us_deaths,
    by = c("Admin2", "Province_State", "Country_Region", "Combined_Key", "date"))
```

Checking new United States data set.

```
head(united_states)
names(global)
```

Joined united_states data set looks good.

Adding Population to Global Data Set.

However, we do not yet have population for the global data set. Adding it below.

```
global <- global %>%
  unite("Combined_Key",
    c(`Province_State`, `Country_Region`),
    sep = ", ",
    na.rm = TRUE,
    remove = FALSE)
global
```

Reading in the file from John Hopkins Github Website. A file with population for countries.

```
population <- read_csv("https://raw.githubusercontent.com/CSSEGISandData/COVID-19/refs/heads/master/csse
  select(-c(Lat, Long_, Combined_Key, code3, iso2, iso3, Admin2))
```

```
## Rows: 4321 Columns: 12
## -- Column specification -----
## Delimiter: ","
## chr (7): iso2, iso3, FIPS, Admin2, Province_State, Country_Region, Combined_Key
## dbl (5): UID, code3, Lat, Long_, Population
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
population
```

Adding population to the global data set.

```
global <- global %>%
  left_join(population, by = c("Province_State", "Country_Region")) %>%
  select(-c(UID, FIPS)) %>%
  select(Province_State, Country_Region, date, cases, deaths, Population, Combined_Key)
global
```

Data Visualization.

United States by State Summaries.

```
US_by_state <- united_states %>%
  group_by(Province_State, Country_Region, date) %>%
  summarize(
    cases = sum(cases, na.rm = TRUE),
    deaths = sum(deaths, na.rm = TRUE),
    Population = sum(Population, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    deaths_per_million = deaths * 1000000 / Population
  )
```

Summary of each state by date.

```
US_by_state
```

Checking the end of the US_by_State data set.

```
tail(US_by_state)
```

OK, we have all the states.

Now look at US totals data set.

```
US_totals <- US_by_state %>%
  group_by(Country_Region, date) %>%
  summarize(
    cases = sum(cases, na.rm = TRUE),
    deaths = sum(deaths, na.rm = TRUE),
    Population = sum(Population, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    deaths_per_million = deaths * 1000000 / Population
  ) %>%
  select(
    Country_Region, date, cases, deaths,
    deaths_per_million, Population
  )
```

```
US_totals
```

Checking the end of the US_totals data set.

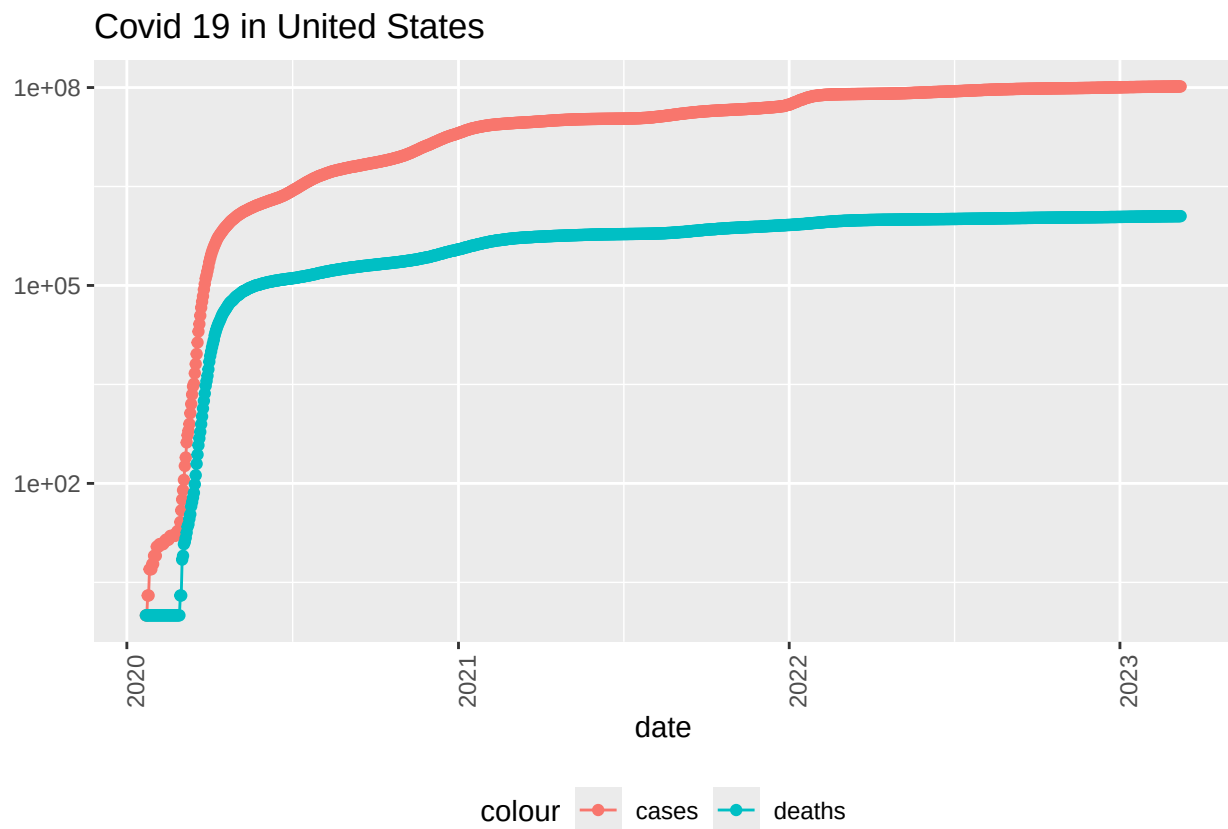
```
tail(US_totals)
```

Data looks good.

Visualization 1

Graph of United States Covid 19 cases and deaths.

```
US_totals %>%
  filter(cases > 0) %>%
  ggplot(aes(x = date, y = cases)) +
  geom_line(aes(color = "cases")) +
  geom_point(aes(color = "cases")) +
  geom_line(aes(y = deaths, color = "deaths")) +
  geom_point(aes(y = deaths, color = "deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom",
        axis.text.x = element_text(angle = 90)) +
  labs(title = "Covid 19 in United States", y = NULL)
```

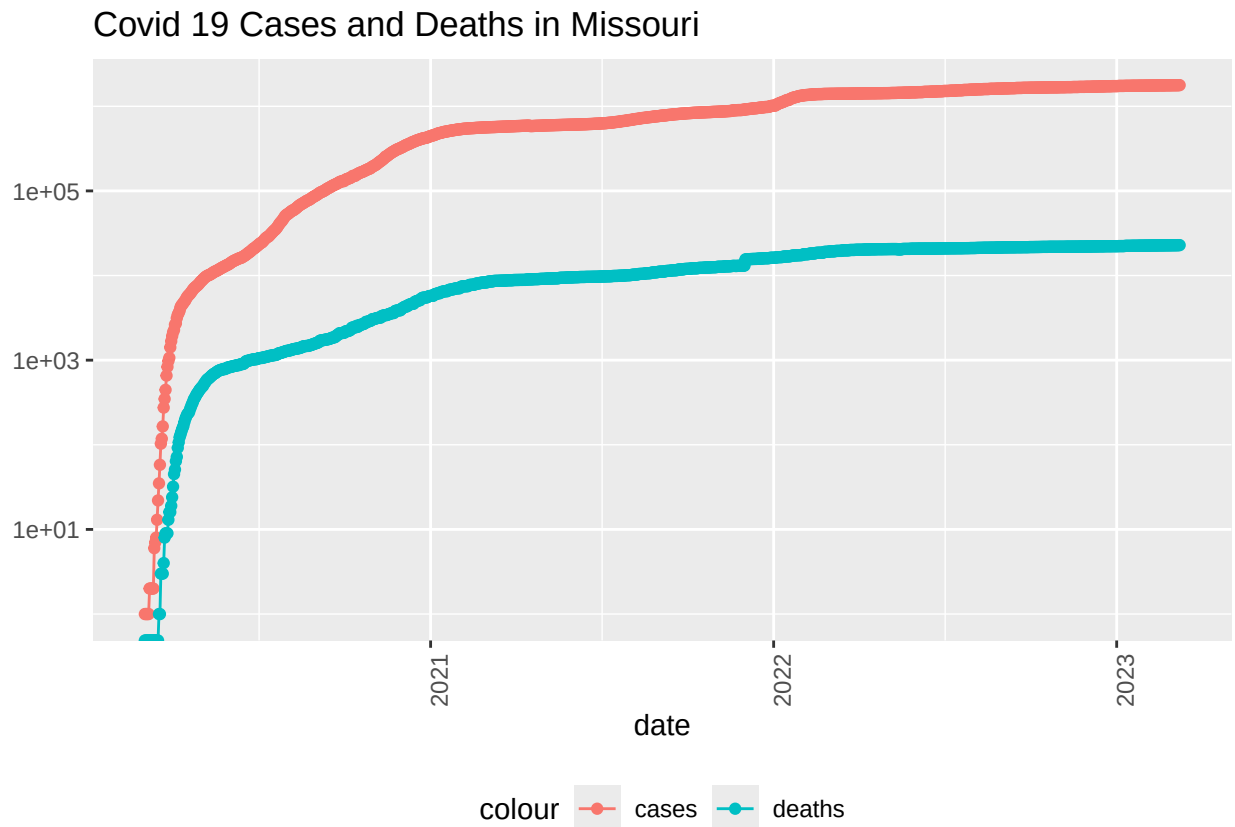


Visualization 2

Graph of Missouri Covid 19 cases and deaths.

```
state <- "Missouri"
US_by_state %>%
  filter(Province_State == state) %>%
  filter(cases > 0) %>%
  ggplot(aes(x = date, y = cases)) +
  geom_line(aes(color = "cases")) +
  geom_point(aes(color = "cases")) +
```

```
geom_line(aes(y = deaths, color = "deaths")) +
geom_point(aes(y = deaths, color = "deaths")) +
scale_y_log10() +
theme(legend.position = "bottom",
      axis.text.x = element_text(angle = 90)) +
labs(title = str_c("Covid 19 Cases and Deaths in ", state), y = NULL)
```



Getting max date and deaths for United States and Missouri.

```
US_max_date <- max(US_totals$date)
US_max_deaths <- max(US_totals$deaths)
```

```
US_max_date
```

```
## [1] "2023-03-09"
```

```
US_max_deaths
```

```
## [1] 1123836
```

Analyzing Data

It looks like the number of cases and deaths taper off and almost flatlines. Is this true or not? I will investigate further.

Adding new variable and transform data again. Adding new_cases variable which is cases - lag(cases). Adding new_deaths variable which is deaths - lag(deaths).

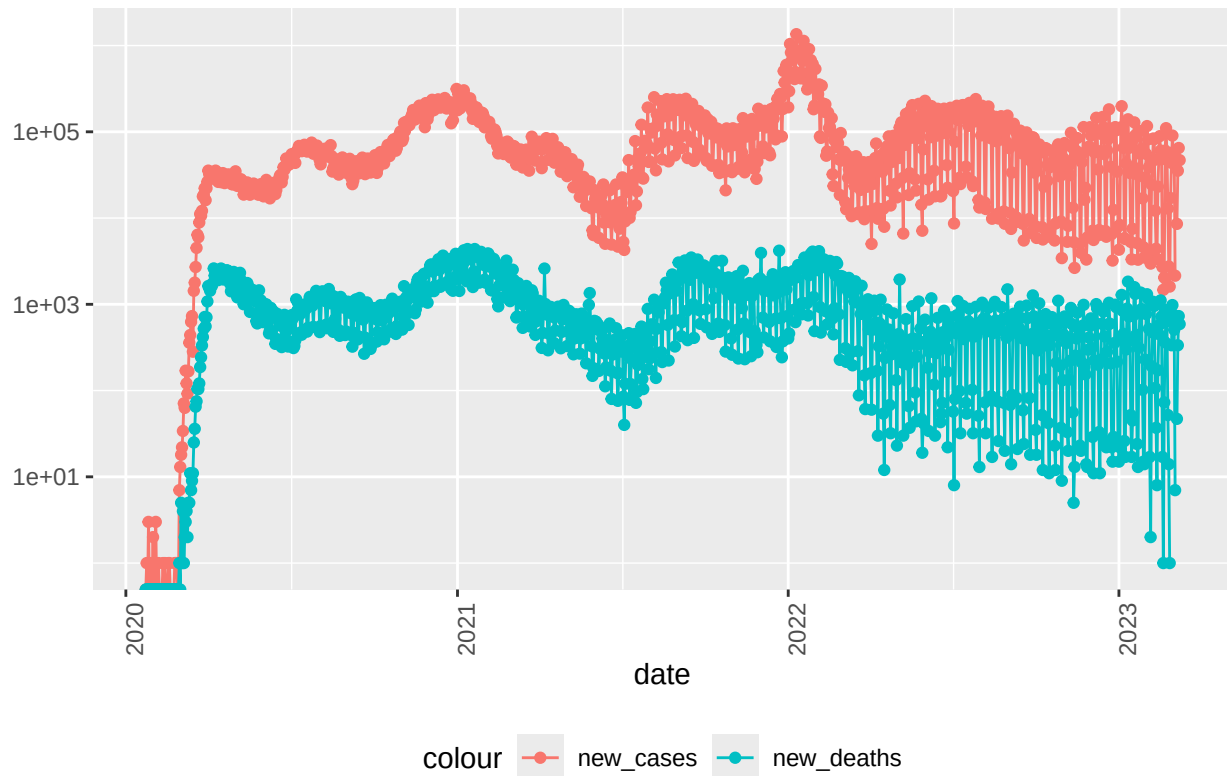
```
US_by_state <- US_by_state %>%
  mutate(new_cases = cases - lag(cases),
         new_deaths = deaths - lag(deaths))
US_totals <- US_totals %>%
  mutate(new_cases = cases - lag(cases),
         new_deaths = deaths - lag(deaths))
tail(US_totals %>%
     select(new_cases, new_deaths, everything()))
```

Visualization 3

Now we can graph new cases and new deaths.

```
US_totals %>%
  ggplot(aes(x = date, y = new_cases)) +
  geom_line(aes(color = "new_cases")) +
  geom_point(aes(color = "new_cases")) +
  geom_line(aes(y = new_deaths, color = "new_deaths")) +
  geom_point(aes(y = new_deaths, color = "new_deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom",
        axis.text.x = element_text(angle = 90)) +
  labs(title = "Covid 19 New Cases and Deaths in United States", y = NULL)
```

Covid 19 New Cases and Deaths in United States



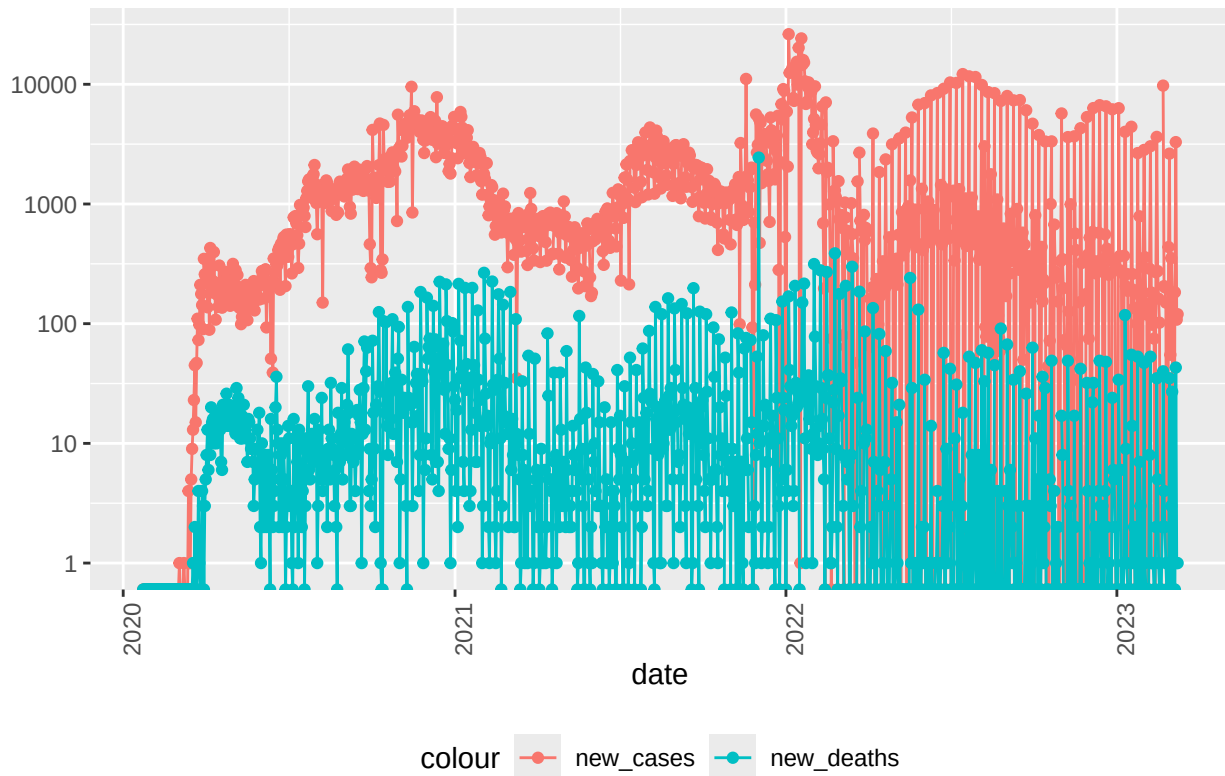
Now we can see the up and down trends in new cases and new deaths in more detail although the overall trend does look like it has flattened out. However, the number of new cases and new deaths remain high.

Visualization 4

Now let's look at the same graph for Missouri.

```
state <- "Missouri"
US_by_state %>%
  filter(Province_State == state) %>%
  ggplot(aes(x = date, y = new_cases)) +
  geom_line(aes(color = "new_cases")) +
  geom_point(aes(color = "new_cases")) +
  geom_line(aes(y = new_deaths, color = "new_deaths")) +
  geom_point(aes(y = new_deaths, color = "new_deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom",
        axis.text.x = element_text(angle = 90)) +
  labs(title = str_c("New Covid 19 Cases and Deaths in ", state), y = NULL)
```

New Covid 19 Cases and Deaths in Missouri



Missouri has a similar pattern of new covid 19 cases and new deaths to the United States totals. There is a dramatic increase in new cases and deaths at the end of 2021 followed by a dramatic decrease in new cases and deaths at the beginning of 2022. After the decrease in cases and deaths at the beginning of 2022 the cases gradually rise again and seem to level off. However, the number of new cases and new deaths remain high.

Modeling Data

Lets look at a linear model of deaths_per_thousand being a function of cases_per_thousand in the US_state_totals data set.

Making US_state_totals data set to use with linear model.

```
US_state_totals <- US_by_state %>%
  group_by(Province_State) %>%
  summarize(deaths = max(deaths), cases = max(cases),
            population = max(Population),
            cases_per_thou = 1000 * cases / population,
            deaths_per_thou = 1000 * cases / population) %>%
  filter(cases > 0, population > 0)
```

Now running the linear model of deaths per thousand as a function of cases per thousand.

```
linear_model <- lm(deaths_per_thou ~ cases_per_thou, data = US_state_totals)
summary(linear_model)
```

```
## Warning in summary.lm(linear_model): essentially perfect fit: summary may be
## unreliable

##
## Call:
## lm(formula = deaths_per_thou ~ cases_per_thou, data = US_state_totals)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.253e-14 -1.946e-15 -7.910e-16  2.950e-16  6.356e-14
##
## Coefficients:
##              Estimate Std. Error    t value Pr(>|t|)
## (Intercept)  -6.077e-14  7.798e-15 -7.792e+00 2.13e-10 ***
## cases_per_thou  1.000e+00  2.497e-17  4.005e+16 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.269e-15 on 54 degrees of freedom
## Multiple R-squared:      1, Adjusted R-squared:      1
## F-statistic: 1.604e+33 on 1 and 54 DF, p-value: < 2.2e-16
```

Highly significant and almost a perfect fit. Further investigation needed to ensure data was cleaned and formatted correctly. This will be done in the future and not part of this visualization and analysis.

Find the number of cases that are the smallest.

```
US_state_totals %>% slice_min(cases_per_thou)
```

Find the number of cases that are the largest.

```
US_state_totals %>% slice_max(cases_per_thou)
```

Cases per thousand go from a minimum of 55641 in American Samoa to a maximum of 460697 in Rhode Island.

Adding prediction column based on the linear model above to the data set.

```
US_state_totals %>%
  mutate(pred = predict(linear_model))
```

From the output the deaths per thousand are almost exactly predicted by the linear model.

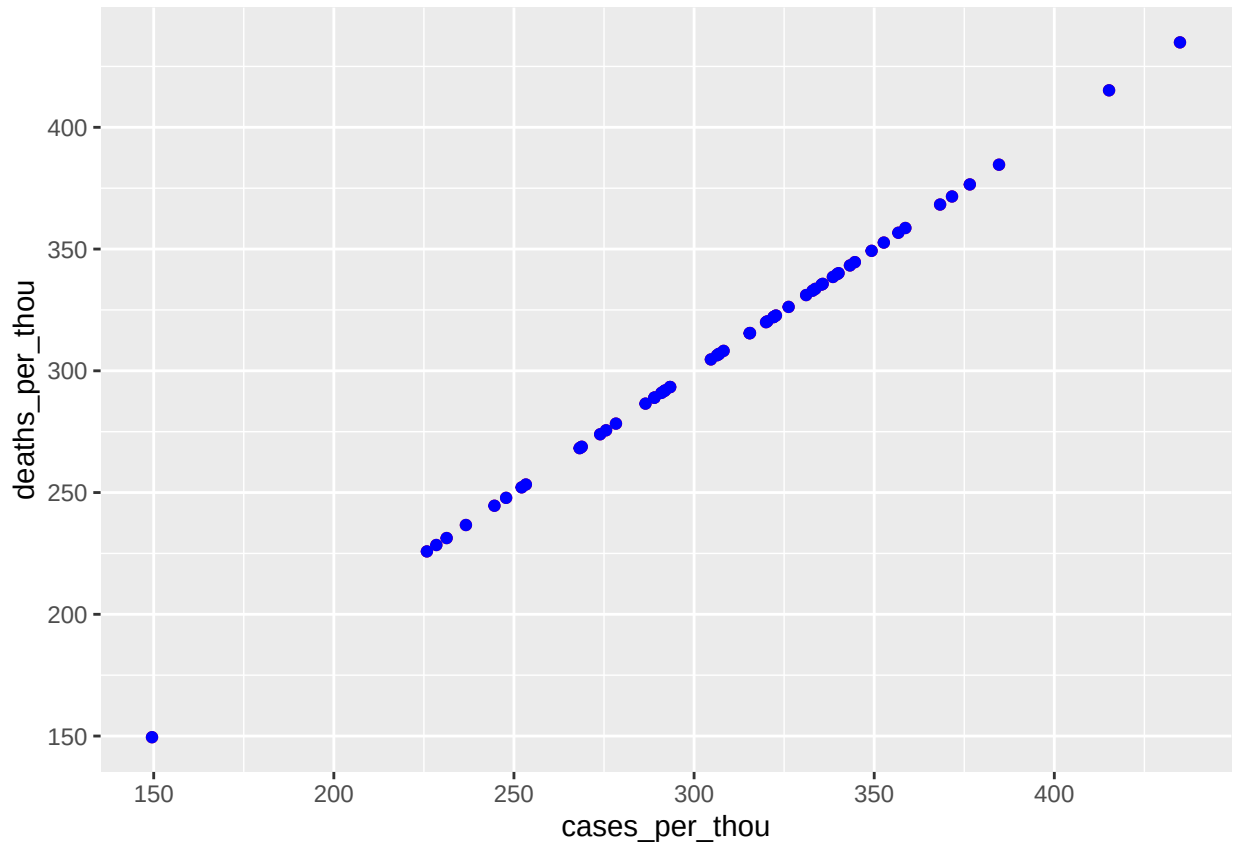
Make new dataset US totals with predictions.

```
US_totals_w_predictions <- US_state_totals %>%
  mutate(pred = predict(linear_model))
US_totals_w_predictions
```

Model Visualization

Plotting actual deaths per thousand vs predicted deaths per thousand.

```
US_totals_w_predictions %>%
  ggplot() +
  geom_point(aes(x = cases_per_thou, y = deaths_per_thou),
             color = "red") +
  geom_point(aes(x = cases_per_thou, y = pred), color = "blue")
```



Conclusion

From the graph, the predicted values almost exactly predict the actual values. This graph along with the results from the linear model run above warrant further investigation of the data to see if an error was made in transforming and cleaning the data because this fit seems too good to be true. Our model says that cases per thousand almost perfectly predicts deaths per thousand in the United States. Further investigation could also look at using this model across countries of the world. Bias in the data could come from the ways in which deaths are reported, do they differ among states and among countries. Also, how quickly the deaths per day are reported. Are they all reported the same day or does reporting time lag in less developed countries.

Session Information

```
sessionInfo()
```

```

## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/Chicago
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] lubridate_1.9.5 forcats_1.0.1  stringr_1.6.0  dplyr_1.2.1
## [5] purrr_1.2.2     readr_2.2.0    tidyr_1.3.2    tibble_3.3.1
## [9] ggplot2_4.0.3   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] bit_4.6.0          gtable_0.3.6      crayon_1.5.3      compiler_4.6.1
## [5] tidyselect_1.2.1   parallel_4.6.1    scales_1.4.0      yaml_2.3.12
## [9] fastmap_1.2.0      R6_2.6.1          labeling_0.4.3    generics_0.1.4
## [13] curl_7.1.0         knitr_1.51        pillar_1.11.1     RColorBrewer_1.1-3
## [17] tzdb_0.5.0         rlang_1.3.0       utf8_1.2.6        stringi_1.8.9
## [21] xfun_0.60          S7_0.2.2          bit64_4.8.2       otel_0.2.0
## [25] timechange_0.4.0   cli_3.6.6         withr_3.0.3       magrittr_2.0.5
## [29] digest_0.6.39      grid_4.6.1        vroom_1.7.1       rstudioapi_0.19.0
## [33] hms_1.1.4          lifecycle_1.0.5   vctrs_0.7.3       evaluate_1.0.5
## [37] glue_1.8.1         farver_2.1.2      rmarkdown_2.31    tools_4.6.1
## [41] pkgconfig_2.0.3    htmltools_0.5.9

```